Undergraduate Biology Students' Climate Change Communication Experiences Indicate a Need for Discipline-Based Education Research on Science Communication Education about Culturally Controversial Science Topics

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ABSTRACT

Science communication is a key skill for undergraduates, but little research explores how biology students communicate about societally important, yet controversial topics like climate change. In this study, we explored whether and how biology students took on the role of science communicators about climate change. We surveyed 191 biology students at 38 universities about their climate change communication frequency and preparedness. We interviewed 25 of the survey participants about their experiences communicating about climate change and their needs when learning about climate change communication. We found that students were communicating about climate change and felt confident discussing the causes and effects of climate change, but they were less confident discussing the solutions to climate change. Students tended to "preach to the choir" by mostly communicating with those who already accepted climate change and avoiding interacting with others who disagreed with them about climate change. Students described a lack of science communication training but had a desire to be taught effective communication skills. Our interviews indicate that if these students felt more prepared to communicate, it may make them more willing to discuss climate change and particularly with people who have different views from them.

INTRODUCTION

Influential reports across science disciplines emphasize the importance of developing undergraduate students' science communication skills (AAMC-HHMI, 2009; AAAS, 2011; ACS, 2015; Joint task force on Undergraduate Physics Programs, 2016; Clemmons *et al.*, 2020), but there is little attention on how this large pool of future scientists communicate about culturally controversial science topics, or topics for which there is scientific consensus but public disagreement (Walker *et al.*, 2017; Bowen *et al.*, 2023). Biology is a discipline that intersects with society in many important, meaningful, and contentious ways, from the safety and efficacy of vaccines to the evolutionary ancestry of humans and human-induced climate change. Whether it be upper-level courses such as immunology that delve into the science of vaccines or introductory classes that often include evolution and ecology, undergraduate biology students are learning about culturally controversial science topics.

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Unsurprisingly then, undergraduate biology students have already been shown to be attempting the act of boundary spanning, or relaying information between the scientific community and their other communities (Hawkins and Rezazade, 2012; Shah *et al.*, 2022). For instance, undergraduate students were significant communicators about COVID-19 vaccines (Couch *et al.*, 2022; Bowen *et al.*, 2023) and have also been communicating about climate change (Shah *et al.*, 2022). But many of these students may be communicating ineffectively. Research shows that during the global COVID-19 pandemic, undergraduate biology students used ineffective tactics to communicate, which included an overemphasis on scientific information and an underemphasis on the socioemotional components of communicating (Couch *et al.*, 2022). If these students are already communicating about culturally controversial science topics, but in ineffective ways, this indicates a need for more research on how to teach communication about these topics. And although we know about the potentially ineffective tactics students used to communicate about COVID-19, there is very little research on how students are communicating about climate change beyond showing that they are doing it (Shah *et al.*, 2022). If we want to improve the current and future discourse on climate change in society, understanding how current undergraduates communicate and what they need to improve their skills is important.

Effective science communication will only become more crucial as the planet continues to warm due to human activities (Pidgeon and Fischhoff, 2011; Maibach *et al.*, 2015; Dixon *et al.*, 2017). Are undergraduate students being prepared to be effective communicators about climate change? In this study, we explore how undergraduate biology students are communicating about climate change and how they have been prepared to do so by their undergraduate curriculum.

BACKGROUND

Impacts of climate change communication

Anthropogenic, or human-induced, climate change has already caused widespread damage to the world's ecosystems and has greatly impacted the health and prosperity of humankind (IPCC, 2014; WHO, 2023). Although scientists have been warning about the dangers of anthropogenic climate change for more than 100 years (Tyndall, 1863; Arrhenius, 1896; Callendar, 1938; Plass, 1956; Maslin, 2016), they have struggled to communicate effectively (McBean and Hengeveld, 2000; Getson *et al.*, 2022). The vast difference between the perceptions of those in the public about climate change and those of climate scientists demonstrates this struggle; while 97% of climate scientists agree that climate change is being caused by humans (Doran and Zimmerman, 2009; Anderegg *et al.*, 2010; Cook *et al.*, 2013), almost half of Americans do not know about this scientific consensus (Leiserowitz *et al.*, 2019). Differences in political and economic values contribute to variations in perceptions about the impacts of climate change (Shwom *et al.*, 2015; Tyson *et al.*, 2023). For instance, 78% of Democrats perceive climate change as a major threat to the United States, while only 23% of Republicans view it as a major threat to the United States (Tyson *et al.*, 2023). Further, despite climate scientists' consistent calls for action against climate change to avoid human health crises (Deng *et al.*, 2020; WHO, 2021; Martin, 2023), 64% of Americans indicated that they rarely or never discuss climate change, and most only hear about climate change in the media once per month or less (Leiserowitz *et al.*, 2019). While the lack of concern and meaningful discussion could have numerous causes besides discrete ineffective tactics from science communicators, such as misinformation, misconceptions, or miscommunication, we have many indications that climate change communication has not effectively illustrated the certainties nor gravity of anthropogenic climate change to the public. The way that climate change science and other culturally controversial science topics are communicated can influence perceptions of these topics and, thus, public support for addressing these issues. For example, discussing free market solutions that emphasize conservative economic values may increase climate change acceptance among those who are politically conservative (Dixon *et al.*, 2017) which could increase public support for climate policy (Kotchen *et al.*, 2013; Sibley and Kurz, 2013) and decrease the negative impacts of climate change. Given that the main goal of education is to improve the well-being of individuals and society (Dewey, 1903), and that communication is a key skill for every undergraduate science student to learn (AAAS, 2011; ACS, 2015; Clemmons *et al.*, 2020), we as biology educators should foster effective communication skills among our future scientists—but how can we accomplish that?

Undergraduate science communication education

Science communication and science education experts note that there is a need for science communication training in undergraduate science courses (AAAS, 2011; Bray *et al.*, 2012; Brownell *et al.*, 2013b; Mercer-Mapstone and Kuchel, 2016; Donohue *et al.*, 2021; Kelp *et al.*, 2024). Science communication is an important skill for undergraduates to develop as it can improve their understanding of science (Mercer-Mapstone and Kuchel, 2016), decision-making regarding science topics (Zohar and Nemet, 2002), and preparedness for success in the workplace (Joint task force on Undergraduate Physics Programs, 2016). Further, science communication training has been connected to increases in students' science identity and science self-efficacy (Alderfer *et al.*, 2023), which in turn can increase their motivation and engagement with science communication behaviors (Murphy and Kelp, 2023). So, when students learn science communication, it may bolster their sense of belonging in the science community as well as make them feel more confident in their ability as scientists. Undergraduate science students also value science communication training. When asked whether science communication should be a topic covered in science classes, 81% of students in chemistry said "yes" and 16% said "maybe" (Donohue *et al.*, 2021). In undergraduate biology specifically, students tend to perceive learning science communication as important and relevant to their lives (Cline *et al.*, 2021).

However, undergraduate biology students may not receive science communication training unless they take extracurricular classes, often outside of their science discipline. While students may get practice with science communication in their biology classes, it is typically communication between themselves and other scientists/science students and not between themselves and diverse audiences outside of their class. Many

studies currently documenting science communication training describe workshops being hosted for professional scientists (Dudo *et al.*, 2021), classes being offered to graduate students (O'Keeffe and Bain, 2018; Wiener and LeFevre, 2022), and extracurricular courses for undergraduate students (Lemus *et al.*, 2014; Laranjo *et al.*, 2020; Osterhage and Rogers-Carpenter, 2022). And while students value science communication training, they perceive time and attention as costs of obtaining it (Cline *et al.*, 2021). So, they may never seek out this training despite valuing it if they have to take an additional course solely dedicated to science communication. In order for the majority of undergraduate science students to receive science communication training, it may need to be embedded within disciplinary content and classes.

Undergraduate science students also report that they desire training on how to communicate about culturally controversial science topics in interpersonal settings, but there is currently very little indication that students are receiving it. Studies on undergraduate students indicate that science communication trainings have no specific focus on controversial topics (Brownell *et al.*, 2013a; Mercer-Mapstone and Kuchel, 2017; Shivni *et al.*, 2021). However, undergraduate students may be more likely to have interpersonal conversations specifically about contentious topics relevant to their lives. For instance, in a study on COVID-19 communication, during the height of the COVID-19 pandemic, undergraduate students reported talking with others outside of class about COVID-19 vaccines often (Couch *et al.*, 2022) and in interviews with students from Black and African American communities, students also reported having interpersonal conversations about COVID-19 vaccines, and although they were taught about the science related to the virus in their classes, they were not taught strategies for effective communication about it (Bowen *et al.*, 2023). These students saw themselves as having a responsibility to communicate effectively and desired training in communication with the aim of lessening the impact of the pandemic on their communities. So, students were participating in interpersonal conversations that they did not feel they were prepared to have. In this study, we are now exploring how students might also engage in discussions on climate change and how their undergraduate science classes may—or may not—have prepared them for these conversations.

Because undergraduate science students are the next generation of scientists and science communicators within their communities, we posit that it is important that they receive science communication training in their disciplinary classes guiding them on effective ways to have interpersonal conversations about culturally controversial science topics. These students come from many diverse communities, including conservative and religious communities, for which there tend to be higher rates of distrust in science (Gauchat, 2015; Simpson and Rios, 2019; Evans and Hargittai, 2020). They may even be the only person in their family with college-level science education (Shah *et al.*, 2022). Thus, they have the potential to be boundary spanners by effectively communicating about scientific topics on a regular basis with people in their communities who may have beliefs counter to scientific consensus (Hawkins and Rezazade, 2012; Shah *et al.*, 2022).

Effective and ineffective science communication strategies

To encourage more effective science communication about culturally controversial science topics, research has explored communication strategies that improve public perceptions of culturally controversial science topics. Recently, experts have worked on transitioning scientists from a deficit approach to dialogue and inclusive approaches when communicating science (Canfield *et al.*, 2020; Vickery *et al.*, 2023). When scientists use a deficit approach, they assume that there are deficits in scientific knowledge and that simply imparting knowledge will be enough to improve science literacy (Nisbet and Scheufele, 2009). However, prior research suggests that transmitting scientific evidence will not be enough to improve perceptions of science if the other person already distrusts the evidence or the messenger (Kubin *et al.*, 2021; Unsworth and Voas, 2021). For example, evolutionary biologist and former Oxford Chair for the Public Understanding of Science, Richard Dawkins, uses a deficit approach to communication about evolution and religion. He is known for his critiques on religion; in his most popular book, *The God Delusion*, he presents religion and evolution as primarily in conflict and criticizes religious beliefs for being, in his opinion, irrational and against scientific evidence (Dawkins, 2006). But this approach causes harm to the public's acceptance of evolution because there are many scientists who have reconciled their religions with evolution (Dobzhansky, 1973; Miller, 1999; Collins, 2006; Graves, 2022) and a high perceived conflict between religion and evolution is one of the biggest factors predicting evolution acceptance (Barnes *et al.*, 2021); it is unsurprising then that knowledge of Dawkins' views on religion was associated with decreased evolution acceptance among a sample of Muslims and Pentecostal Christians (Unsworth and Voas, 2021).

Science communication experts have recommended that instead of focusing on deficits in scientific knowledge and making negative assumptions of the audience's intelligence, scientists should engage in a two-way dialogue in which information is an exchange rather than a one-way transmittance (Borchelt and Hudson, 2008). Using this dialogue approach, scientists and their audiences can discuss their perspectives and concerns about the science topic in a meaningful way that leads to mutual knowledge gain (Borchelt and Hudson, 2008; Mercer-Mapstone and Kuchel, 2017). Scientists can also be more effective in their communication by using more inclusive approaches, such as by focusing on the diversity of the audience and potential factors that may negatively impact their engagement and access to science (Canfield *et al.*, 2020). One study showed that when scientists have a deficit perspective of the public about science, it presents a barrier to them engaging in dialogue and inclusive science communication practices (Choi *et al.*, 2023). By respecting the audience's backgrounds, acknowledging the cultural context surrounding science topics, and including diverse perspectives, science communication can be more effective for fostering trust and engagement with science (Bray *et al.*, 2012; Mercer-Mapstone and Kuchel, 2017; Canfield *et al.*, 2020).

Current undergraduate science communication

Based on prior research, we know that undergraduate science students communicate about culturally controversial

science topics with others, but, like current scientists, they may be using ineffective strategies (Couch *et al.*, 2022; Shah *et al.*, 2022; Bowen *et al.*, 2023). In an interview study exploring first-generation undergraduate students' science communication, students described attempting to be boundary spanners when communicating about science to their communities (Shah *et al.*, 2022). Some students mentioned that they communicated about culturally controversial science topics, including climate change, but that these topics were particularly challenging to discuss (Shah *et al.*, 2022). However, this study was broadly focused on students' science communication, and not necessarily just about controversial topics. In studies about COVID-19 communication, undergraduate students were communicating frequently about COVID-19 vaccines at a crucial time for effective science communication when the vaccines were first becoming available to the public (Couch *et al.*, 2022; Bowen *et al.*, 2023). However, when these students were asked to describe how they would respond to others who were hesitant to receive a vaccine, the most prevalent approaches that students described was a deficit approach in which they *only* imparted scientific information, such as statistics, measurements, and research results to their audiences.

Undergraduate biology students were also serving as significant communicators about COVID-19 vaccines among communities that had increased vulnerability to COVID-19 (Bowen *et al.*, 2023). Interviews exploring Black undergraduate science students' COVID-19 communication revealed that these students often described themselves as boundary spanners and trusted messengers in Black communities about COVID-19 vaccines, where COVID-19 had a disproportionate impact (Muñoz-Price *et al.*, 2020; Mackey *et al.*, 2021), but they used a limited number of strategies recommended by science communication scholars (Bowen *et al.*, 2023). The results from these studies indicate that undergraduate science students could be important science communicators and could benefit from undergraduate science communication education. However, prior studies have not focused on students' experiences communicating about anthropogenic climate change, which is another culturally controversial science topic; while experts agree that the climate is changing due to human activity, there is significant disagreement among the public. We don't know how frequently students discuss climate change with their friends, family, and others in person and on social media, which has become one of the dominant forms of communication among young adults. By understanding what topics students are already communicating about, we can better tailor science communication training to topics that are relevant to current events and their lives more broadly. We also do not know the extent to which students are prepared to communicate about climate change. Given that the science of climate change is not a standard core component of undergraduate biology education, biology students may not have a well-developed knowledge of climate change, yet they still may be talking about it. If students do not feel prepared to communicate about climate change, we can incorporate climate change education and communication training in undergraduate courses to better prepare them to be science communicators within their communities.

Theoretical Frameworks: Boundary spanners and cultural competence

To explore how undergraduate science students communicate to others about climate change, we use the theoretical framework of boundary spanning. Boundary spanning involves the greater ability of an individual to facilitate an exchange of information between two different groups due to their unique belonging and experiences within both groups (Hawkins and Rezazade, 2012). Boundary spanners are often trusted messengers between groups. For example, Black scientists are important science communicators in Black communities because they can bridge potential communication gaps between their communities and scientific institutions and build trust in science (Pietri *et al.*, 2018; Alsan *et al.*, 2021). In the context of science communication, first-generation undergraduate science students have exhibited boundary spanning roles by translating scientific information to their communities in a way that is more inclusive of the diverse knowledge levels and the range of viewpoints in those communities (Shah *et al.*, 2022). We can further explore potential boundary spanning roles in undergraduate science students by studying their *climate change* communication specifically. Are students engaging with communities where there may be variability in knowledge levels of, or trust in the scientific evidence for, climate change?

We also used the framework of cultural competence. While boundary spanning acknowledges how *similarities* between people make communicating across boundaries *easier*, cultural competence recognizes how *differences* between people make communication across boundaries more *difficult* (Betancourt *et al.*, 2003; Tanner and Allen, 2007; Barnes and Brownell, 2017). When one becomes more culturally competent, they become more effective communicators with those who have different cultures or beliefs than themselves. For instance, we have previously used this framework to describe effective communication between mostly secular biologists and mostly religious students while teaching evolution, another culturally controversial science topic that has strong support from scientists but less support from some religious communities (Barnes and Brownell, 2017; Truong *et al.*, 2018). We used this framework in our current study to describe how students being enculturated into science may communicate with people who are not a part of the scientific community and whom they have different identities or cultures that make communicating effectively more difficult. Thus, undergraduate science students may benefit from using culturally competent practices when communicating with their communities about topics such as climate change. For instance, students can find shared values with those they are communicating with despite them possibly coming from different backgrounds. Indeed, using value-based messaging when communicating about climate change can be an effective way of improving conservatives' climate change perceptions (Dixon *et al.*, 2017). If students use culturally competent practices when communicating about culturally controversial science topics, they may be more effective communicators and seen as more credible.

Current study and research questions

The goal of this study is to explore undergraduate science students' communication about climate change and how they feel

either prepared or unprepared to have these conversations, particularly with those who they may disagree with. We also aim to understand whether and how students learned about climate change communication in their science classes and how they can become more prepared to communicate effectively. The research questions we asked fell within two broad categories:

Communication experiences and preparedness: What are undergraduate biology students' experiences communicating about climate change within their communities, and how prepared do they feel to communicate effectively with those who have differing views from themselves?
Learning experiences and needs: What are undergraduate biology students' experiences learning about climate change communication, and what are their perceived needs?

MATERIALS AND METHODS

Middle Tennessee State University's Institutional Review Board approved this study (protocol no. 22-20267).

Recruitment

We recruited a national sample of undergraduate biology students from research universities to participate in this study. In Fall 2021, we sent a recruitment message to biology department heads at 260 private and public research-intensive universities across the United States. The list of schools came from the Carnegie classification list (Carnegie Classification, 2022); the emails for the department heads were found on the department website for each school. The recruitment email asked the department heads to share a message asking their undergraduate students to volunteer for a survey about their climate change communication experiences in exchange for a chance to win a $50 gift card. We used the survey to collect quantitative data on their communication frequency and preparedness and qualitative data on their strategies they use to communicate. The survey respondents also served as a sample pool for interviews. We sent an email to survey participants who agreed with items on a climate change acceptance survey and who indicated they communicated about climate change more than once a year inviting them to participate in an interview in exchange for a $25 gift card. We only selected students who accepted anthropogenic climate change because this study focuses on students who communicate the scientific consensus view of anthropogenic climate change and not on those who communicate rejection of anthropogenic climate change. We selected students who communicated about climate change at least once a year because we wanted to hear from students who had experiences communicating.

Surveys

We designed a survey to explore variables related to students' climate change communication as well as to create a pool of participants for potential interviews. The survey collected data on how often students communicated about climate change and how prepared they felt to communicate accurately about different aspects of climate change. The survey asked students about the extent of their communication about climate change, who they discussed climate change with, how frequently they communicated, and how they communicated about climate change on social media. We adapted these questions from a survey used in a prior study exploring biology students' communication about COVID-19 (Couch *et al.*, 2022). To understand the frequency of students' climate change communication, we asked, "How frequently do you talk about climate change with others?" (never, about once a year, about once a month, about once per week, more than once per week). We were also interested in seeing who students were discussing climate change with and what percentage of the time they were talking with their family, friends, or others about climate change. Additionally, we wanted to understand whether and how students were communicating about climate change on social media since this is a popular form of communication among young adults that can influence their willingness to engage in communication about the environment (Knupfer *et al.*, 2023). To do this, we asked about the frequency in which students saw, interacted with, or wrote posts on social media about climate change (never, about once a year, about once a month, about once per week, more than once per week). To select students for interviews who accepted climate change, we used a previously published instrument comprised of six questions (i.e., "The climate is always changing and what we are currently observing is just natural fluctuation" and "The burning of fossil fuels over the past 50 years has caused serious damage to the planet's climate") on a 5-point Likert scale (Dixson *et al.*, 2017). To build response process (substantive) validity evidence for our measures (AERA, 2014), we conducted cognitive interviews (Castillo-Diaz and Padilla, 2013) with 5 undergraduate students to revise any language in the surveys that was unclear to students.

To understand how students would communicate with people who have views on climate change different from theirs, we asked students what they would say to a friend or family member who did not accept anthropogenic climate change. We analyzed these open-ended survey responses from students who agreed with items on the climate change acceptance survey ($n = 125$). We developed a coding rubric using inductive coding and constant comparison methods (Glaser and Strauss, 2017; Glesne and Peshkin, 1992; Cho and Lee, 2014; Krippendorff, 2019). Two researchers (C.D.B and M.G.A.) developed independent codebooks after reading all of the survey responses. C.D.B. compared these codebooks and combined them to create a single codebook. To establish interrater reliability of the codebook, two researchers (A.R.S. and C.D.B) used it to independently code a subset of 10% of responses, and we calculated the Cohen's κ interrater score and deemed it acceptable ($\kappa = 0.83$; Landis and Koch, 1977). Both researchers coded all responses, and disagreements were discussed and resolved except in the case of 3/191 responses for which disagreement between the two researchers was persistent and, in these cases, a third researcher (M.E.B.), decided the final code. We used the prior research literature and published expert recommendations to categorize students' strategies as effective or ineffective (e.g., Bray *et al.*, 2012; Mercer-Mapstone and Kuchel, 2017; Aurbach *et al.*, 2019; Wack *et al.*, 2021)

Study context

To contextualize the sample, we collected the students' demographic information, including their gender, race/ethnicity, political identity, year in school, major, and intended career as part of our survey. We collected information about students' political identities to illustrate the cultural diversity of the sample and their potential to be boundary spanners. In total, 191 students from 38 universities and 21 states completed the survey, and we interviewed 25 undergraduate students about their experiences communicating about climate change. The majority (70%) of institutions that respondents attended were public universities, with equal representation of institutions classified as high and very high research activity (Carnegie Classification, 2022). 3.6% of students were first generation (neither parent had received a bachelor's degree or higher), and the majority of our sample were domestic students. Students in this study were at a wide range of points in their academic careers. 17% of students identified as being in their first year of study, 17% were in their second year, 26% were in their third year of study, and 40% were in their fourth year of study or beyond. 83% of the students who responded to the survey identified majoring in biology or something biology-related, including but not limited to botany, environmental science, pre-health, and ecology, and 47% of all students intended on pursuing careers in healthcare.

The aggregated demographics of the 191 survey participants are in Table 1. All survey questions analyzed for this paper and the coding rubric used to analyze students' open-ended responses can be found in the Supplemental Materials.

Table 1. Demographic breakdown of students in this study.

		Survey % ($n = 191$)
Gender	Man	16.0
	Woman	79.05
	Nonbinary	4.0
	Decline to state	0.5
Race/ethnicity	White	62.0
	Black/African American	6.0
	Asian	14.0
	Latinx or Hispanic	7.0
	Multiracial	8.05
	Native	1.0
	Other race/ethnicity	0.5
	Decline to state	0.5
Religion	Christian	42.4
	Muslim	3.7
	Hindu	3.7
	Jewish	2.6
	Buddhist	1.6
	Other religion	3.7
	Not religious	38.7
	Decline to state	3.7
Political Identity	Liberal	60.7
	Not liberal	28.3
	Decline to state	11.0
Year	First year	17.0
	Sophomore	17.0
	Junior	26.0
	Senior	36.0
	Decline to state	4.0
Major	Biology	83.0
	Non-Biology	13.0
	Decline to state	4.0
Career	Research Scientist	24.0
	Healthcare Professional	47.0
	Other career	28.0
	Decline to state	1.0

Interviews

To explore students' actual experiences communicating, we developed an interview script to explore students' experiences communicating about climate change in detail and their current habits and strategies used when conversing with people with differing views. We asked students to describe their personal communication experiences with a particular focus on conversations with people who disagreed with them and to describe their perceived preparedness and effectiveness in those conversations. We also wanted to explore their prior preparation in biology classes for communicating about climate change with those who may not accept anthropogenic climate change, so we asked students to describe times when they were taught how to communicate to them about climate change. We also asked students to describe what they thought they needed to get better at communicating about climate change. In this study, we focused on exploring the ways students tried to build trust by using relational strategies because sharing research findings is unlikely to be compelling in the absence of trust. So, during this exploratory research with students, we did not aim to probe and describe the actual research findings students communicated, but the relational strategies they used or did not use. A copy of the final interview script is provided in the Supplemental Materials. These semistructured interviews were audio recorded, and transcribed via Zoom, and were an average of 45 min long. A single researcher (M.G.A.) conducted all interviews to ensure consistency across interviews.

Interview analyses

After the interview, the interviewer took detailed notes to begin to identify patterns. We conducted interviews past data saturation in which no new themes emerged (Guest *et al.*, 2006). After all interviews were completed, they were transcribed and pseudonyms were assigned to each interview transcript to protect the participants' identities. Because pseudonyms were assigned and decoupled from the survey data, we do not know the demographic information of the interview participants. The transcripts were analyzed using inductive coding methods with constant comparisons (Krippendorff, 2019; Cho and Lee, 2014; Glaser and Strauss, 2017; Glesne and Peshkin, 1992) to identify themes that emerged in the interview data. The interviewer (M.G.A.) and the principal investigator on the project (M.E.B.) developed a preliminary set of codes based on the 25 interviews conducted. Then, two additional researchers (A.R.S. and C.D.B.) independently reviewed all interview transcripts to identify any additional themes. We iteratively developed a codebook based on students' climate change communication experiences and strategies by continuing this process until

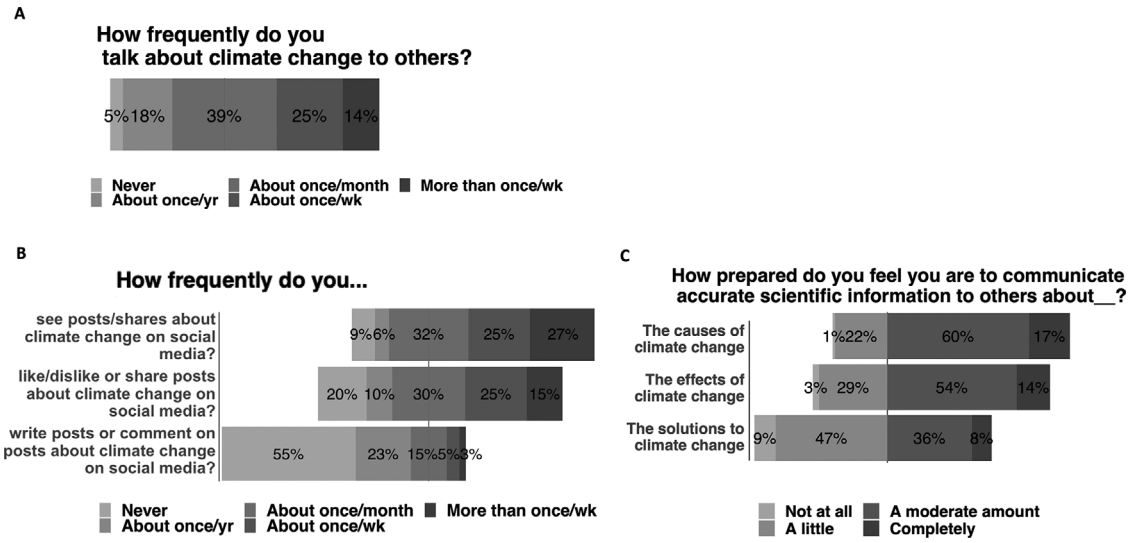


Figure 1. Students' self-reported climate change (A) communication frequency, (B) communication via social media, and (C) communication preparedness.

the four researchers agreed on a set of themes in the data. We completed an interrater reliability analysis by having two researchers independently code five interviews (A.R.S. and C.D.B.) and then comparing the codes each researcher found, which resulted in a high Cohen's κ interrater score (0.90) (Landis and Koch, 1977). C.D.B then used the final codebook to code all interviews. The final coding rubric can be found in the Supplemental Materials. Quotes have been lightly edited to ensure clarity and protect the participants and their institutions. It is important to note that reported theme percentages cannot be generalized to the entire population of undergraduate biology students.

RESULTS

Survey finding 1: Students were communicating about climate change, but they often did not feel prepared to communicate about the *solutions* to climate change.

Communication frequency and preparedness. In our study, we found that undergraduate biology students often acted as science communicators about climate change (Figure 1). 39% of surveyed students reported that they communicated about climate change with others about once a month and another 39% reported that they communicated about climate change at least once a week. Meanwhile, 18% reported communicating about climate change just once a year, and only 5% reported never communicating about climate change. Students reported primarily communicating about climate change with their friends (52%) and family (32%) and rarely with others (16%), indicating that students primarily communicate with those they are close to. In addition, students reported feeling unprepared to communicate about solutions to climate change with others. While the majority of students felt at least moderately comfortable communicating accurately about the causes and effects of climate change, less than half (44%) of students reported feeling at least moderately comfortable communicating accurately about the solutions to climate change (Figure 1).

We also asked about students' communication frequency on social media. Students reported seeing posts about climate change at least once a week on social media, but they did not frequently write or comment about climate change on social media. 55% of students reported never writing or commenting on posts about climate change on social media, and 23% of students reported writing or commenting on posts just once a year (Figure 1).

These findings suggest that students were communicating about climate change, but they felt more prepared to discuss causes and effects than solutions. Additionally, they often saw posts about climate change on social media, but they rarely commented on them or wrote their own. Next, we report the strategies that students said they would use to communicate about climate change.

Survey finding 2: Students described ineffective and effective strategies they would use to communicate with people of differing views than them

We asked students to describe how they would communicate to a friend or family member who did not believe that humans are causing climate change and who was not concerned about climate change. Students described that they would use strategies that can be effective such as communicating local or noticeable evidence of climate change (Scannell and Gifford, 2013; van der Linden *et al.*, 2015) (29%), listening or asking questions to understand the other person (Lewandowsky *et al.*, 2021; Wack *et al.*, 2021) (16%) and consensus messaging (2.4%) (Sauer *et al.*, 2021). However, students also described many strategies that are recommended against such as only providing scientific information (55.3%) (MacDonald *et al.*, 2020); invalidating communication such as telling the other person that their opinion is wrong (9.6%), that they should do something different (15.2%), or being accusatory (9.6%) (Greville-Harris *et al.*, 2016); and avoiding climate change conversations when the other person disagrees with them (4.8%) (Hocker and Wilmot, 2018, pp. 33–34). Students also described tactics that research has not explored or is

Table 2. Themes in students' written responses when they were asked how they would communicate with a friend or family member if the friend or family member said that humans were not causing climate change or they were not concerned about climate change.

Theme	Description of theme	% (n = 125)	Example of student response	Example of student response
Social duty	Student tells the person they have a responsibility to act against climate change	82.4	"This is our only planet and it is important to keep it safe for our children and their children."	"Think about the future of those who will live after us."
General information	Student presents general scientific information	55.4	"The amount of greenhouse gases used has significantly increased since the industrial revolution and has almost a direct correlation to rises in temperature."	"I would pull up facts and data about the proof that there is an effect being made by humans."
Local/noticeable evidence	Student presents local, noticeable, or personalized information	29.3	"…things that are occurring now are not normal like the wildfires and the droughts and the floods."	"…ask them if they had noticed any changes in the local climate over the past decade…These are from climate change I would point out."
Listening/Questions	Student describes listening or asking questions to determine points or acknowledge opinions.	16.0	"I would ask them why they think that and respond accordingly based on their answer."	"I would first ask what reasons they have to believe that in order to make my responses more specific to their thought processes."
Prescribe	Student tells the person they should do something different	15.2	"…look at current data and to educate themselves."	"We should not just be concerned, but actively doing things to combat [climate change]."
You are wrong	Student tells the person they are wrong	9.6	"I would just tell them their comment is obviously not true."	"I'd tell them that it's just not true."
Accusatory	Student accuses person of doing something wrong.	9.6	"They were illogical…"	"It is selfish and naive to not be concerned about climate change."
Depends	Student's communication would vary depending on certain factors.	4.8	"…it depends on how open minded they are towards the topic."	"If I was feeling more apt to [argue], I would make sure that they understand it as a concept."
Avoidance	Student describes avoiding/disengaging from the discussion/person	4.8	"I don't dare go into further debate [about climate change]."	"Admittedly, I am not one to argue with someone over something that has turned political…it can sometimes feel like talking to a brick wall."
Consensus messaging	Student mentions using the scientific consensus on climate change	2.4	"Almost all scientists agree that humans have a strong impact on climate change."	"The vast majority of climate scientists would disagree with you."

unclear whether is effective or ineffective such as explaining that the other person has a social responsibility to act against climate change (82.4%). See Table 2 for descriptions of themes and examples of student responses.

Although our survey gathered some of students' potential climate change communication strategies, we also wanted to explore students' actual communication experiences in more depth and also explore their experiences learning about climate change communication in depth via interviews. Below are the results from these interviews.

Interview findings

To explore students' actual communication experiences within their communities, we interviewed students about their climate change communication and learning experiences that may not have been captured in their survey responses. These data were primarily aimed at describing the landscape of students' experiences rather than quantifying the prevalence of those experiences. Instead of attempting to generalize our findings, we aim to highlight the possible transferability of our findings from our sample to other contexts within the field of biology education. To this end, we aimed to provide as much context as possible about both the students and the universities they represented while maintaining their anonymity (see Study Context), and we aimed to provide detailed descriptions of our participants' responses, since providing context and the quality of description allow for the reader to better evaluate how our findings can be transferred from our context to other institutions or biology students more broadly (Lincoln and Guba, 1985, pp. 298, 316). Given the provided context and the diverse range of institutions and students represented in our study, we do posit that the more prevalent a theme was among participants, the more likely it is to be transferable to other contexts. Particularly, the results are more likely to be transferable to those students similar to the sample in this study, who accept anthropogenic climate change and who would be willing to interview about their communication experiences, which may indicate a higher level of interest in the topic than the biology student population in general. Thus, in the text, we indicate when "most" students (two-thirds or

more), "many" students (between one-third and two-thirds), or "some" students (less than one-third) reported on a theme. We first report communication strategies that students used within their communities that are known to be ineffective based on prior literature. Next, we describe communication strategies that students used that are known to be effective. Finally, we report students' experiences in undergraduate classes learning about climate change communication and what they think they need to be more effective communicators about climate change. All themes that emerged from the data and the coding rubric used to analyze the data can be found in the Supplemental Material.

Student communication experiences and strategies: Preaching to the choir, avoiding, and feelings of frustration

Many students in our study mentioned that most of their boundary-spanning attempts regarding climate change occurred with others who already accept and are concerned about it. They rarely described communicating with people who they knew disagreed with them about climate change. For example, Ava said, "Usually my talk about climate change is with people that agree with me." When communicating with others who accept climate change, some students said that they were discussing climate change mitigation. For instance, Dan mentioned that he communicated frequently with his friends about "solutions that are being proposed for climate change." During these conversations, some students such as Amy described feelings of hopelessness:

> "When we talk about [climate change] …it's always with dread because it's such a huge issue and we kind of feel very helpless about it."—Amy

Many students also described communicating about climate change on social media. This usually involved sharing posts or commenting under posts. For example, Andi mentioned:

> "On social media I say it's more like advocating and just posting… like on Earth Day, I know, a lot of people will post stuff… in general I'm posting stuff like that to spread more education and information about climate change."—Andi

When students did have conversations about climate change with people they disagreed with, they often reported that these contentious conversations were with family members. For example, Claire said,

> "For family [communicating about climate change] is more like starting from ground zero where it doesn't exist, so it's a really different conversation […] my family is typically pretty anti-science and so having conversations revolving around science in any degree is typically a pretty aggressive conversation."—Claire

Further, another student, Chandi, described that during her contentious conversations with family, she was "dismissed and not taken seriously". Miranda described that when she has contentious conversations with her family, it makes her "not hopeful for the change that should be made regarding the issue, because if they refuse to even believe it exists, then how are we going to improve the situation?". So, students described having more positive attempts at boundary spanning when they talked with their friends and others who agreed with them, but they tended to have more negative conversations with family who often don't share the same stance as them.

Students reported using strategies that they knew were ineffective. Some students said that they felt ineffective when they became angry or frustrated during climate change discussions, which is in line with prior literature showing that statements of anger are more likely to evoke antagonistic emotions and behavior and less likely to evoke compassionate emotions or conciliatory responses (Kubany *et al.*, 1992). For example, Jane mentioned, "Whenever it's more ineffective would be when you get more upset and then you just kind of don't think about it as purely."

Riley further described how her frustration impacted her communication:

> "Sometimes it's easy to just get mad. Like why don't you care? This is something that is a big deal…It's just an easy thing to do because you're so frustrated about this social stagnation, in terms of addressing climate change that when you meet an individual person who is bought into the idea that there's nothing we can do and it's not real, it's just very easy to get personally angry at them instead of being an effective communicator."—Riley

Some students even mentioned disengaging from being boundary spanners with people who disagree with them. They described a desire to avoid potential conflict by avoiding these discussions completely. For instance, Jade said, "Sometimes I won't even bring it up with people who wouldn't agree with me just to avoid any uncomfortable feelings." However, completely avoiding any conversations with those who disagree may shut down opportunities for shared understanding and effective communication (Hocker and Wilmot, 2018, pp. 33–34). Alternatively, students may wisely decide to "pick their battles" and not "waste time and energy" on those with extreme views who are unwilling to have a respectful dialogue (Leiserowitz *et al.*, 2021). For instance, another student, Katie explained that even though she feels that avoiding conversations is ineffective, she still feels like it is sometimes practical:

> "I feel like it's definitely [ineffective] to just not communicate, but like I said…it's almost like talking to a brick wall and I just kind of like shut down and I'm like you know what, I'm not even going to tell them the facts that I know or the things that I've learned because it's just not going to get to them."—Katie

In line with recommendations that science communicators should be knowledgeable and establish credibility to be effective (Hendriks *et al.*, 2016), many students explained that they felt ineffective when they were unknowledgeable about climate change topics. One student explained:

> "I definitely don't feel completely confident because I always feel like I don't have the knowledge to bring out…I never feel fully competent in that degree."—Claire

This indicates that teaching science content in ways that are inclusive, accessible, and increase student confidence in their knowledge could be important for establishing effective science communicators.

Students also reported using ineffective strategies that they thought were effective. Some students in our study mentioned that they felt like effective communicators when they provided information about climate change. For instance, Hari said, "What I like to do is mention facts and use statistics and I feel like that's the best way to effectively communicate on my stance". Another student, Ashley, further described that having "passion and then confidence in the facts and making sure you have the right facts" makes her feel effective. She and other students like Nathan further described how being charismatic when presenting information about climate change is important for their communication:

> "There are just a few instances where I felt like it was effective because I was well educated on a certain topic or there is something about climate change I was just passionate about that would make me want to talk about it more to somebody. So, it gave me an optimistic attitude."—Nathan

These quotes indicate that students primarily thought about providing facts and being passionate instead of incorporating more trust-building communication strategies first was an effective way to discuss climate change, even though research shows, and experts recommend, that providing information alone without attending to the socioemotional aspects of the conversation will likely be ineffective, particularly with people who already have high distrust in science (Hart and Nisbet, 2012; Kubin *et al.*, 2021).

These findings suggest that while students were communicating about climate change, they often only communicated with those who also agreed with them, which would indicate they have not yet fostered cultural competence mindsets in which they can have conversations that "reach across the aisle," but they did sometimes have contentious conversations about it when discussing with family members. Students also described feeling hopeless during conversations with those who also accept climate change because they ruminate together about people who do not accept climate change or don't believe that it is a concern. They described feeling ineffective when they lacked knowledge about certain climate change issues, when they became angry, and when they avoided communicating to people who disagreed with them. Students believed they were effective when providing scientific information about climate change even though providing information alone is not recommended because it is less effective than when in conjunction with other strategies (Schweizer *et al.*, 2009). In fact, some evidence supports that providing scientific information alone can increase polarization about controversial science topics (MacDonald *et al.*, 2020). See Table 3 for descriptions of themes and students' responses.

Student communication experiences and strategies: Respect, understanding, and personal relevance

Students reported effective strategies that they knew were effective. Although many students mentioned that they avoided people who they disagreed with and sometimes felt frustrated, some students described experiences when they felt effective as boundary spanners when they were respectful and understanding:

> "Usually when [communicating] with somebody who just doesn't believe that climate change exists…I would pretty much try to understand their side because it's always important to hear somebody out completely before making your counter argument and I think that's something that's really important."—Hari

Here, Hari recognized the importance of active listening when communicating about controversial science topics like climate change (Lewandowsky *et al.*, 2021; Wack *et al.*, 2021). In addition, some students said they felt effective when they were communicating in a way that is easy for the other person to understand. For instance, Samantha mentioned, "If I'm talking to someone who isn't as educated about [climate change], I have to explain all of those terms or use different terms to work around it". Samantha recognized that communicating about climate change using layman's terms rather than scientific terminology would be more effective (Joslyn *et al.*, 2023).

Many students also recognized that they were more effective communicators when they were making the conversation more personal and relevant to the other person (Scannell and Gifford, 2013). Hari mentioned, "I think it's always best to relate to things that are relevant in everybody's lives…how it will impact them directly." For instance, students described specific examples of local and noticeable evidence of climate change that would be most impactful:

> "I would try to show them more basic examples of how it affects more of their day-to-day life. Especially here in New Jersey, we have hurricane season…In the recent years the hurricanes have been a lot stronger, so I just try to relate it to their everyday life or something that they're experiencing."—Brooke

So, students described being effective boundary spanners when using strategies that were more positive. Given that using local and personally relevant examples are published, recommended strategies for effective communication about climate change (Scannell and Gifford, 2013), this reflects cultural competence in that they are using strategies with people of differing characteristics that are shown to be effective (Tanner and Allen, 2007).

These findings indicate that students recognized when they were using effective communication strategies. Students felt effective as boundary spanners when they were communicating in a respectful manner, communicating in a way that the audience can understand, and when they were more personal. See Table 4 for descriptions of themes and students' responses.

Next, we describe how students experienced learning about climate change communication in their undergraduate science classes.

Table 3. Examples of themes from interviews related to experiences of undergraduate biology students communicating about climate change.

Themes	Description	Example Quote	% ($n = 25$)
Limited network	Most of the students' network is limited to people concerned about climate change	"Most people I'm surrounded by definitely have similar opinions to mine."	56%
Avoid disagreement	Does not communicate with people who do not agree with them.	"If I hear that someone doesn't believe in climate change, I tend to avoid the conversation."	52%
Unknowledgeable	Student felt ineffective in communicating when they were not knowledgeable about climate change or people's misconceptions	"Sometimes when I'm not fully educated on the topic …then I feel like I don't communicate as well."	44%
Mitigation discussions	Student discusses with friends and family who are accepting of climate change about how they can mitigate climate change	"My friends have had many discussions about what to do and what in the future can be done to improve this (climate change)."	40%
Contentious conversations with family	Student had contentious discussions about climate change with family	"[Conversations with family] basically end up getting really defensive."	40%
Social media	Student shares and comments under posts about climate change on social media.	"If I see [posts] about pollution…I've always tried to share it."	36%
Provide facts	Providing evidence for or facts about climate change will be effective for communication	"The way to communicate to them is bring up facts…that probably would be the best."	36%
Hopelessness	Student feels hopeless about the implications of climate change when discussing climate change.	"The topic itself is kind of…you feel kind of hopeless about the situation."	24%
Anger/Frustration	Student feels ineffective when they get angry or frustrated while communicating about climate change	"I definitely feel like I've had ineffective conversations with my family about [climate change] because I get so emotional."	24%
Charismatic	Being charismatic by having confidence and passion when communicating is effective.	"I consider myself an effective speaker on something I'm passionate about, so I tried to make myself as compelling as possible."	12%

Student experiences learning about climate change communication: Students often learn about the science of climate change but not how to communicate about it

Most students in our study reported that in classes where they learned about the science of climate change, they were never taught how to communicate about climate change effectively. For example, Ava mentioned that in classes where she did learn about the science of climate change, the "focus was on understanding [climate change] and not trying to communicate." Another student, Claire, further discussed how she felt

Table 4. Examples of themes from interviews about students' effective climate change communication strategies.

Themes	Description	Example Quote	% ($n = 25$)
Economic and personal	Student feels effective when discussing the economic impacts, personal impacts, and local evidence of climate change	"I think that making the conversation applicable to everybody and how it affects everybody is a lot better."	48%
Respect/Understanding	Student says they approach communicating by respecting, understanding, or being nice in the conversation.	"I think presenting it in a non-confrontational way has really helped me become an effective communicator."	32%
Meet them where they are	Student feels effective when using words that a lay person understands	"If you're using a lot of big words and scientific jargon it's going to turn people away."	16%

Table 5. Examples of themes from interviews describing experiences undergraduate biology students had learning about climate change communication.

Themes	Description	Example Quote	% ($n = 25$)
Taught science not communication	Student says they were taught scientific information about climate change but not how to communicate to nonscientists.	"[Courses] have talked about the ozone layer and the science of fossil fuels…but never in a way where I was competent enough to articulate it."	80%
Research/class presentations	Presentations/research for courses have improved their confidence communicating about climate change.	"Researching more in classes and just learning more in the news has improved my communication about climate change."	48%
Personal research	Student's information about climate change and climate change communication came from personal research	"I have a lot of articles that are readable to non-scientists…which is always helpful if you're talking to someone who isn't a scientist."	28%

about not being taught effective climate change communication strategies:

> "We learned about facts, but we haven't really learned how to effectively communicate them. There's definitely a gap in there, where they teach you about [the science of climate change], but they don't teach you about how to have those conversations."—Claire

Claire is describing how there is a gap in training on how to be effective during one-on-one conversations.

Many students mentioned times where doing research and presentations in class made them more confident science communicators to other scientists. For instance, Andi explained, "When I took environmental courses… presenting that research in front of other people kind of made me a better speaker and more confident presenting in front of people".

Because most students were not learning about science communication in their classes, some students sought out information about effective communication through personal research. For instance, Amy explained that she primarily got information about effective science communication from "research projects learning about how to communicate scientific language and concepts." Another student, Chris, mentioned that his education about effective communication came from "personal research… I've never had an actual class here that talks about scientific communication to people who are not in the field." So, students may be recognizing that they may need to learn more than scientific information to be more effective communicators.

These findings indicate that students were learning about the science of climate change in their science classes, but they were not learning how to communicate it effectively with those who may doubt the scientific evidence that shows anthropogenic climate change. Some students sought out personal research to learn how to communicate effectively or through class projects. See Table 5 for descriptions of themes and examples of students' responses.

In the final finding, we describe the communication training and specific practices that students desired to be more effective science communicators.

Undergraduate student needs in order to be more effective communicators about climate change

Many students said that learning more about the science of climate change would help them be more confident communicating with others about climate change. For instance, Jane said, "If I had more knowledge that I used more frequently, I would be more prepared to discuss [climate change]." This aligns with our previous finding where students said that they felt ineffective when they were unknowledgeable about climate change.

All students in this study mentioned that they would benefit from explicit training on how to communicate about climate change more effectively. For example, Katie explained, "…a specific subunit on the actual communication skills would help so much". Students like Ava mentioned that receiving science communication training would be relevant to their career paths:

> "I think it would be good to take a course [that incorporates science communication training] because even in the position that I am right now, not really knowing where I want to go, I feel like everything is related to [climate change]…So I feel like it's really important for any field area that I want to potentially be able to talk about this, because everything is connected."—Ava

Most students also mentioned specific practices that they thought instructors could implement that would improve their confidence communicating about climate change and other contentious science topics. For example, Claire mentioned that "giving examples about conversations that may happen and how to prepare for them" is one thing she thinks instructors should include in their classes. Another student, Hari, suggested, "Maybe have a seminar course, discussion-based forum, where students just read articles and discuss on a weekly basis. I think that'd be an effective way to increase…communication skills".

Some students mentioned that practicing more science communication would help them feel more confident. For example, Tyler described:

Table 6. Examples of themes from interviews regarding what undergraduates think they need to effectively communicate about climate change.

Themes	Description	Example Quote	% ($n = 25$)
Communication training	Student says they would benefit from explicit training on how to communicate climate change to non-scientists.	"I would have rather taken a science communication course because that would have been so much more useful"	100%
Specific practices	Student describes specific practices that would be beneficial in the classroom	"I think instructors need to facilitate more peer-to-peer interactions about this type of communication."	72%
Learning information	Student says learning more information about climate change will help them feel more confident communicating about climate change	"If I could just be more informed, I think I would be able to communicate more effectively."	36%
Practice	Having practice communicating about climate change would make students feel more confident communicating about climate change	"Instead of just sitting down and listening to somebody talk about how to communicate, you have to actually practice."	20%

> "I feel like that's (climate change communication) something that should be more like a practical training, like you actually have to do it. Instead of just sitting down and listening to somebody talk about how to communicate, you have to actually practice."—Tyler

Another student, Anne further described, "I would say just more practice about it. No one ever really brings up communication about [climate change], they bring up what it is, but they don't bring up how to talk about things that are so controversial". Students like Tyler and Anne recognized that they needed more than just knowledge about the science of climate change, they also needed practice communicating about contentious science topics with those who may disagree with them.

These results indicate that students want communication training in their science courses to help them become more confident science communicators. Many students described that they needed to learn more about the science of climate change to help them feel more confident, but all students mentioned that they would benefit from science communication training. Most students described specific practices that instructors could implement in their science courses to help students communicate effectively. The themes are summarized in Table 6.

DISCUSSION

In this study, we found that students were attempting to be boundary spanners in their communities, but they may need more support in their undergraduate education to become more culturally competent when communicating about climate change with people that they disagree with. While students we interviewed described using some culturally competent strategies in their communication, they also discussed using behaviors and tactics that would not be culturally competent with those of different backgrounds. Below we summarize the strategies students were using, whether they align with effective science communication recommendations and research, and how we might develop future research to support science communication education in undergraduate science classes.

Students' use of effective or ineffective communication strategies may help or hinder their efforts to be boundary spanners and culturally competent communicators

Undergraduate biology students described using, or that they would potentially use, communication strategies that are known to be effective based on prior literature. Students described being respectful and understanding, which is an important and culturally competent strategy for communicating with others that they disagree with. Being respectful of the audience's views can foster trust in the scientist and thus the information being communicated (Bray *et al.*, 2012). Similarly, some students described that they would listen to the other person or ask them questions regarding their climate change hesitancies. Actively listening to the audience's concerns can promote empathy and trustworthiness of the communicator (Goodwin and Dahlstrom, 2014; Lewandowsky *et al.*, 2021). In addition, some students mentioned that they meet the other person where they are by, for example, reducing the use of scientific jargon, which has been shown to be effective because it allows the audience to have a better understanding of science (Mercer-Mapstone and Kuchel, 2017). Some students mentioned that they would bring up the scientific consensus about climate change, which has evidence for improving climate change attitudes (Ding *et al.*, 2011; Lewandowsky *et al.*, 2013; van der Linden *et al.*, 2017). Finally, in both the survey and interviews, students recognized that discussing the local or noticeable evidence and personal impacts of climate change can be an effective communication strategy. Emphasizing information about climate change in a way that the audience perceives to be more relevant has led to positive climate change attitudes (Myers *et al.*, 2012; Dixon *et al.*, 2017).

However, students also described using communication strategies that are known to be ineffective and may pose barriers to boundary spanning and cultural competence. The most prevalent strategy that students described using was to only

impart scientific information to the other person. This indicates that most students in this study may have been taking a deficit approach to communication like we have seen in prior studies (Couch *et al.*, 2022; Bowen *et al.*, 2023). Some students also described that they would use invalidating communication by telling the other person to do something different or accusing them of doing something wrong, further suggesting that students may be taking a deficit approach when communicating. These strategies that focus on assumed deficits in the audience's knowledge have been shown to have limited effectiveness (Unsworth and Voas, 2021) and could even result in less accepting climate change beliefs. Students correctly recognized that being angry and frustrated when communicating with someone who has differing views than them was an ineffective strategy. When discussions become hostile, communication becomes less effective (Kubany *et al.*, 1992). To avoid potential hostility, about half of the interview sample described hesitating to have conversations with people who have differing views than them and instead communicated with like-minded individuals. The fact that students reported communicating about climate change with those they disagree with aligns with what we have seen in prior studies indicating that students can be potential boundary spanners in their communities (Shah *et al.*, 2022; Bowen *et al.*, 2023). However, avoiding these conversations can be a negative strategy because it limits the potential to engage in meaningful dialogue with others about climate change (Burns *et al.*, 2003; Van der Sanden and Meijman, 2008; Mercer-Mapstone and Kuchel, 2017). On the other hand, avoiding conversations with those who have extreme views may be adaptive because those with extreme views are unlikely to change their minds and may be more motivated toward creating conflict than resolutions (Leiserowitz *et al.*, 2021).

In addition to these strategies, students also described other methods that lack evidence of effectiveness. For example, the most prevalent strategy students described is to tell the other person that they have a social responsibility to act against climate change. Although studies have focused on the impacts of discussing the public health benefits of vaccine communication (Betsch *et al.*, 2017) and climate change communication (Myers *et al.*, 2012), there is limited peer-reviewed research showing evidence for the effectiveness of describing one's social responsibility to act against climate change when communicating. Students also mentioned that when they discuss climate change with others who also agree with their climate change beliefs, they talk about feeling hopeless about people who don't believe that climate change is a concern. It is possible that emphasizing shared frustrations can have *negative* effects on subsequent interactions with outgroup members, such as inflaming intergroup conflict with the targeted group (Myers, 1975) and reducing students' cultural competence with members of groups that differ from themselves. Additionally, lamenting with community members about shared frustrations with out-group members could have *positive* effects by providing social support in times of stress and fostering greater support within the group (Jetten *et al.*, 2014), which could increase boundary spanning behaviors within their immediate cultural groups with people they know share their views on climate change. For instance, a community member might accept that humans are causing climate change but not accept that their individual actions can help mitigate climate change. A science student lamenting about their shared frustrations of "climate change deniers" may induce social trust with this person and increase the likelihood they would receive communication that individual actions can make a difference. However, especially given the potential negative effects of intergroup conflict increases, we recommend that students increase their boundary spanning without using shared frustrations with outgroups in their communication.

Implications for science communication education

Undergraduate biology students that we surveyed were frequently communicating with others about climate change but did not feel prepared to communicate about the solutions to climate change. Researchers note that incorporating solutions and hope more in climate change communication can encourage engagement and action (Ojala, 2012; Dearing and Lapinski, 2020), so students should feel confident explaining potential solutions to climate change if we want them to be better communicators. Further, students mentioned feeling ineffective in their communication when they were unknowledgeable about certain climate change topics, and many students said that gaining more knowledge about climate change would help them feel more prepared to communicate effectively. Therefore, teaching about the science of climate change often and making sure that students feel confident in their knowledge is likely important (Nam and Ito, 2011; Flora *et al.*, 2014; Kumar *et al.*, 2023). The communicator feeling confident and knowledgeable and presenting as an expert is important for effective communication (André *et al.*, 2011), which was in line with student responses. So, one important step to increasing students' climate change communication skills is to foster climate literacy (National Oceanic and Atmospheric Administration, n.d.) in undergraduate education. Undergraduates have been shown to harbor basic misconceptions about climate change such as conflating climate change with ozone depletion and misattributing the causes of climate change (Wachholz *et al.*, 2014; Huxster *et al.*, 2015). However, there is a distinct lack of broad recommendations for what undergraduates should learn about climate change in higher education settings such as biology classes. To make better science communicators, we may need to have clearer content and communication goals for climate change education in undergraduate biology. For content, we can look to K-12 guidelines that are further along in being established to inform undergraduate guidelines. Across research studies and published guidelines for K-12 there are common themes in suggestions for climate change education such as teaching about the nature of climate science, sources and impacts of greenhouse gases, the role humans play in causing climate change, the threats of climate change to humans and ecosystems, climate change mitigation, climate action, and climate justice (see Table 7 for citations). However, K-12 guidelines generally do not have suggestions for teaching science communication about climate change and given that our research is finding that undergraduates are already significant communicators about this topic, express a desire to improve their communication, yet rarely receive it, that we should consider including goals for teaching about

Table 7. List of common themes in K-12 climate change education guidelines and the supporting literature.

Theme	Citations
Teaching about the nature of climate science	(Johnson, 2011; Climate Literacy and Energy Awareness Network, 2016; National Center for Science Education, n.d.)
Sources and impacts of greenhouse gases	(Warny and Lindstedt, 2016; Climate Literacy and Energy Awareness Network, 2016; New Jersey State Department of Environmental Protection, 2023)
Role humans play in causing climate change	(Climate Literacy and Energy Awareness Network, 2016; New Jersey State Department of Environmental Protection, 2023)
Threats of climate change to humans and ecosystems	(Johnson, 2011; Climate Literacy and Energy Awareness Network, 2016; New Jersey State Department of Environmental Protection, 2023)
Climate change action and mitigation strategies	(Johnson, 2011; Climate Literacy and Energy Awareness Network, 2016; National Center for Science Education, n.d.; North American Association for Environmental Education, 2022; United States Department of Agriculture, n.d.)
Climate justice	(Climate Literacy and Energy Awareness Network, 2016; North American Association for Environmental Education, 2022; New Jersey State Department of Environmental Protection, 2023; Milks *et al.*, 2024)

climate change communication when teaching about climate change science.

Increasing climate literacy is likely necessary but insufficient to create effective science communicators. All students in our study recognized that they would be more effective communicators if they received climate change communication training in their classes and described that they were only taught concepts but not science communication in their classes. Instructors could teach students communication skills beyond content knowledge, such as culturally competent practices, to help students become more effective communicators. However, there is very little guidance for how instructors should go about teaching science communication about controversial topics in undergraduate science classes because undergraduate science communication education has not often been developed in discipline-specific settings like undergraduate biology (for exceptions see: Mercer-Mapstone and Kuchel, 2016, 2017; Wack *et al.*, 2021). We also need more evidence-based strategies for teaching communication and disciplinary content simultaneously in science classes to allow students to communicate more effectively about climate science and the impacts of climate change.

LIMITATIONS AND FUTURE RESEARCH

Our findings indicate a need for improved science communication education in undergraduate science courses. Specifically, future research should focus on developing more instruments and measures to evaluate the effectiveness of science communication education about culturally controversial science topics, including important outcomes such as communication knowledge, skill, and self-efficacy. There is also a need for more instructional frameworks of best practices for implementing undergraduate science communication education. With more frameworks, more materials can be developed and evaluated to explore potential improvements in students' communication outcomes. Future research is also needed to examine other science communication strategies that have limited evidence for effectiveness to determine which practices instructors should include when teaching about science communication.

One potential limitation of our study is that we did not recruit students from community colleges or primarily undergraduate institutions. Future research is needed to explore the communication experiences of community college students because they are rarely studied (Schinske *et al.*, 2017; Barnes *et al.*, 2022). In addition, students volunteered to take a survey about a controversial topic with limited incentive and thus may have been interested in climate change communication. So, we might also need to look at the communication experiences of students who are not already interested in climate change.

Another potential limitation of this study is that we did not explore students' knowledge of climate change. Since it is not included as one of the core concepts of biology (AAAS, 2011) and there are no broad recommendations for undergraduate climate change education, there were no expectations that students would have expertise on climate change. Additionally, in their interviews, students also frequently mentioned providing "facts" without defining what, in their eyes, constitutes a fact about climate change. Researchers often discuss conflation of terms such as fact, theory, and law as part of a broader discussion of the nature of science (Lederman and Lederman, 2014, pp. 600–617; McComas and Clough, 2020). Students themselves may not know what they mean by a "scientific fact". However, we did not explore this area of student conceptions about the nature of a scientific fact.

We recognize the importance of understanding how much students are learning about climate change and future research should explore to what extent students are receiving content knowledge about climate change in their courses. If students' climate change knowledge is minimal and does not extend beyond the conception that climate change is occurring and that humans are accelerating the process, then students need considerably more content knowledge to be able to better understand the science behind climate change and to be able to communicate about it more effectively.

CONCLUSION

In this study, we found that undergraduate biology students were communicating about climate change with others but described feeling unprepared to communicate about the solutions to climate change. They described experiences when they felt effective and ineffective at communicating. They also described communicating the most with friends, family, and

people who hold similar views on climate change, but avoided conversations with people who hold differing views. Although students reported learning about the science of climate change in their undergraduate classes, they did not learn strategies for communicating about climate change effectively and expressed a desire for climate change communication training in their classes to help them feel more confident as communicators. This study contributes to the current biology education *and* science communication literature by specifically exploring the climate change communication and learning experiences of undergraduates. Our findings imply that students are already climate change communicators, which could have significant implications for our ability to address climate change as a society. This adds to the accumulating body of literature accumulating that shows undergraduates are significant communicators about culturally controversial science topics. Thus, to address the needs of our students and society, we need to build a more comprehensive research foundation in biology education on the needs of undergraduates for advancing their skills and confidence in science communication about controversial topics.

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